```
use ig_clone;
select * from users;
select * from photos;
select * from follows;
select * from comments;
select * from likes;
select * from tags;
select * from photo_tags;

# A. Marketing Analysis
# 1. Loyal User Rewards:- Five oldest users on Instagram

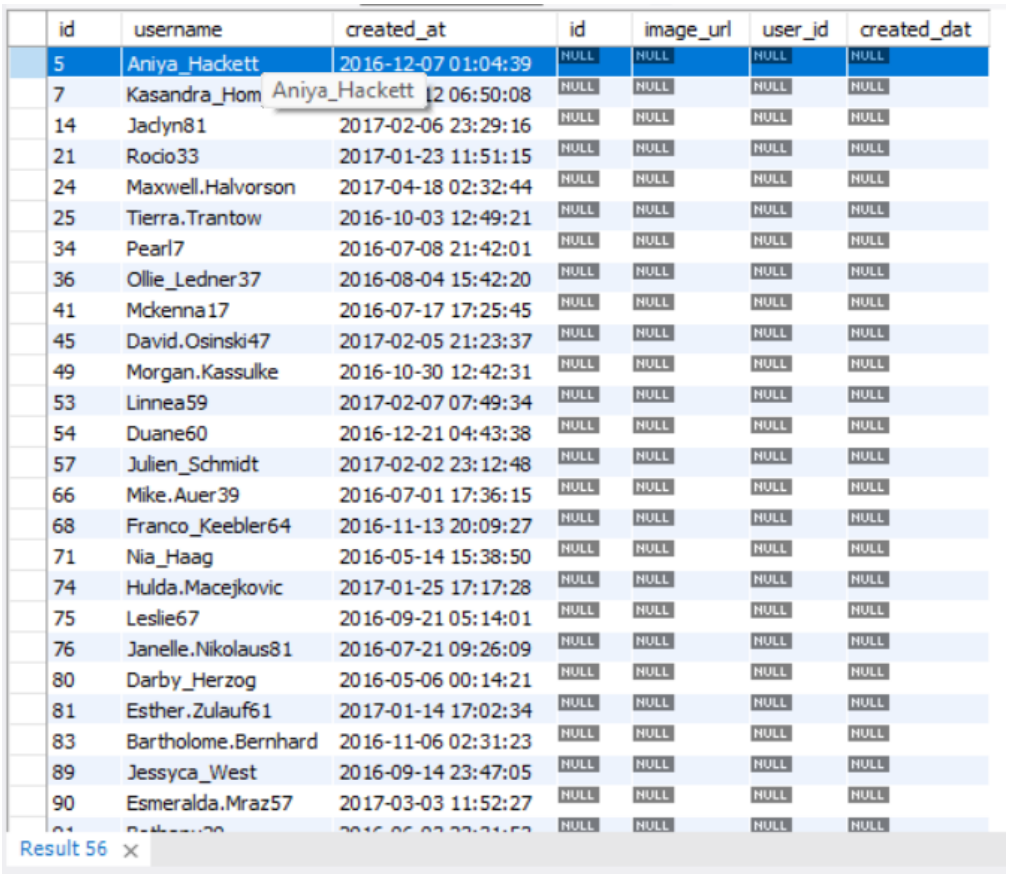
select * from users;
SELECT
   username, created_at
FROM
   users
ORDER BY created_at
LIMIT 5;
```

username	created_at
Darby_Herzog	2016-05-06 00:14:21
Emilio_Bernier52	2016-05-06 13:04:30
Elenor88	2016-05-08 01:30:41
Nicole71	2016-05-09 17:30:22
Jordyn.Jacobson2	2016-05-14 07:56:26

```
# 2. Inactive User Engagement:- User never posted photo

select * from photos;
SELECT
   *
```

```
FROM
    users
        LEFT JOIN
    photos ON photos.user_id = users.id
WHERE
    photos.image_url IS NULL;
```

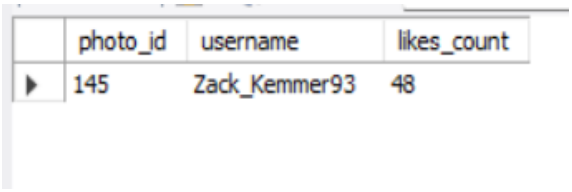


id	username	created_at	id	image_url	user_id	created_dat
5	Aniya_Hackett	2016-12-07 01:04:39	NULL	NULL	NULL	NULL
7	Kasandra_Hom	12 06:50:08	NULL	NULL	NULL	NULL
14	Jaclyn81	2017-02-06 23:29:16	NULL	NULL	NULL	NULL
21	Rocio33	2017-01-23 11:51:15	NULL	NULL	NULL	NULL
24	Maxwell.Halvorson	2017-04-18 02:32:44	NULL	NULL	NULL	NULL
25	Tierra.Trantow	2016-10-03 12:49:21	NULL	NULL	NULL	NULL
34	Pearl7	2016-07-08 21:42:01	NULL	NULL	NULL	NULL
36	Ollie_Ledner37	2016-08-04 15:42:20	NULL	NULL	NULL	NULL
41	Mckenna17	2016-07-17 17:25:45	NULL	NULL	NULL	NULL
45	David.Osinski47	2017-02-05 21:23:37	NULL	NULL	NULL	NULL
49	Morgan.Kassulke	2016-10-30 12:42:31	NULL	NULL	NULL	NULL
53	Linnea59	2017-02-07 07:49:34	NULL	NULL	NULL	NULL
54	Duane60	2016-12-21 04:43:38	NULL	NULL	NULL	NULL
57	Julien_Schmidt	2017-02-02 23:12:48	NULL	NULL	NULL	NULL
66	Mike.Auer39	2016-07-01 17:36:15	NULL	NULL	NULL	NULL
68	Franco_Keebler64	2016-11-13 20:09:27	NULL	NULL	NULL	NULL
71	Nia_Haag	2016-05-14 15:38:50	NULL	NULL	NULL	NULL
74	Hulda.Macejkovic	2017-01-25 17:17:28	NULL	NULL	NULL	NULL
75	Leslie67	2016-09-21 05:14:01	NULL	NULL	NULL	NULL
76	Janelle.Nikolaus81	2016-07-21 09:26:09	NULL	NULL	NULL	NULL
80	Darby_Herzog	2016-05-06 00:14:21	NULL	NULL	NULL	NULL
81	Esther.Zulauf61	2017-01-14 17:02:34	NULL	NULL	NULL	NULL
83	Bartholome.Bernhard	2016-11-06 02:31:23	NULL	NULL	NULL	NULL
89	Jessyca_West	2016-09-14 23:47:05	NULL	NULL	NULL	NULL
90	Esmeralda.Mraz57	2017-03-03 11:52:27	NULL	NULL	NULL	NULL

3. Context Winner Declaration:- User with most like on a single photo

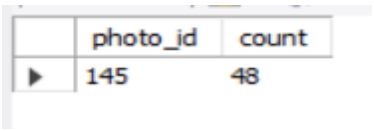
```
SELECT
    likes.photo_id,
    users.username,
    COUNT(likes.user_id) AS likes_count
FROM
    likes
```

```
    INNER JOIN
  photos ON likes.photo_id = photos.id
    INNER JOIN
  users ON photos.user_id = users.id
GROUP BY likes.photo_id , users.username
ORDER BY likes_count DESC
LIMIT 1;
```



photo_id	username	likes_count
145	Zack_Kemmer93	48

```
SELECT
  photo_id, COUNT(*) AS count
FROM
  likes
GROUP BY photo_id
ORDER BY count DESC
LIMIT 1;
```

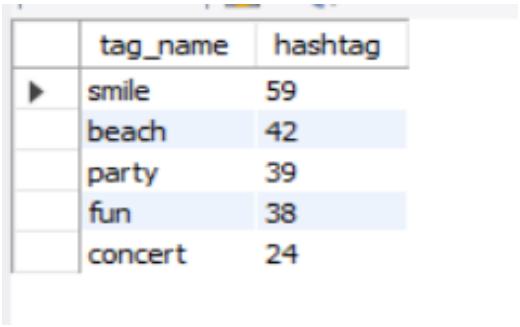


photo_id	count
145	48

4. Hashtag Research:- Top 5 hashtag use in their post

```
select * from photo_tags, tags;
SELECT
  tag_name, COUNT(photo_id) AS hashtag
FROM
  photo_tags
    INNER JOIN
  tags ON tags.id = photo_tags.tag_id
```

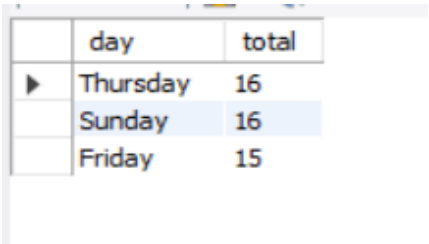
```
GROUP BY tags.tag_name
ORDER BY hashtag DESC
LIMIT 5;
```



tag_name	hashtag
smile	59
beach	42
party	39
fun	38
concert	24

5. AD Campaign Launch:- Best day to launch ads

```
SELECT
   DAYNAME(created_at) AS day, COUNT(*) AS total
FROM
   users
GROUP BY day
ORDER BY total DESC
LIMIT 3;
```



day	total
Thursday	16
Sunday	16
Friday	15

B. Investors Metrics

1. User Engagement:- average number of posts per user and the total number of photos divided by the total number of users.

```
SELECT
```

```
    ROUND((SELECT
                COUNT(*)
            FROM
                photos) / (SELECT
                COUNT(*)
            FROM
                users),
        2) AS Avg_post;
```



Avg_post
2.57

2. Bots and Fake Account:- users (potential bots) who have liked every single photo on the site, as this is not typically possible for a normal user.

```
with CTE as(
SELECT
    username, COUNT(photo_id) AS likes
FROM
    likes
        INNER JOIN
    users ON users.id = likes.user_id
GROUP BY users.username
)
SELECT
    username, likes
FROM
    CTE
WHERE
    likes = (SELECT
```

```
    COUNT(*)
FROM
    photos);
```

username	likes
Aniya_Hackett	257
Jadyn81	257
Rocio33	257
Maxwell.Halvorson	257
Ollie_Ledner37	257
Mckenna17	257
Duane60	257
Julien_Schmidt	257
Mike.Auer39	257
Nia_Haag	257
Leslie67	257
Janelle.Nikolaus81	257
Bethany20	257